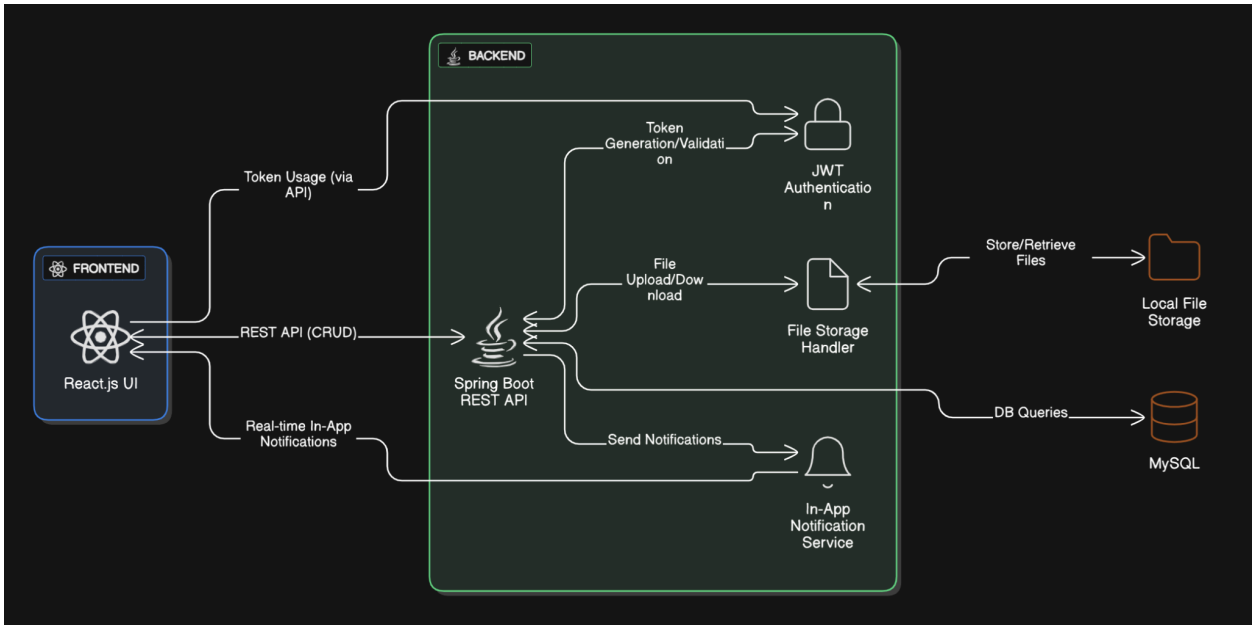
Study Group Finder & Collaboration Platform

Project Statement:
The **Study Group Finder & Collaboration Platform** is a web application that helps students connect with peers in the same courses to form effective study groups. Users can create profiles, list their enrolled courses, discover others taking the same subjects, and collaborate via built-in communication and productivity tools. The system promotes academic networking, improves study efficiency, and makes group work coordination seamless.

Outcomes:

- User authentication and profile creation with basic academic details.
- Add and manage a list of enrolled courses.
- Search and discover other students in the same course.
- Create and join study groups with group name, description, and privacy settings.
- Group chat for real-time communication.
- Integrated calendar for scheduling study sessions.
- Notifications for upcoming sessions, new group invites.

Architecture diagram:



Modules:

- **Module A:** User authentication and course management
- **Module B:** Group creation, discovery, and membership management
- **Module C:** Group chat and shared document editing

- **Module D:** Calendar scheduling and notifications

Milestones:

Milestone 1: Week 1–2 — Authentication & Course Management

- Implement user registration/login with JWT-based authentication.
- Profile setup with name, academic details, and avatar upload.
- Add/remove enrolled courses from a predefined course list.
- Display a dashboard of all joined groups and suggested peers.

Outcomes:

- Login and registration forms
- Profile editing form
- Dashboard page

Milestone 2: Week 3–4 — Group Creation & Discovery

- Create study groups with course association, description, and privacy options.
- Search and filter groups by course, size, or activity.
- Request to join private groups or instantly join public ones.
- Group member list

Outcome:

- Create group form.
- Search group with filters

Milestone 3: Week 5–6 — Communication & Collaboration

- Real-time group chat using WebSockets.

Outcome:

- Chat page.
- Messaging widget

Milestone 4: Week 7–8 — Scheduling & Notifications

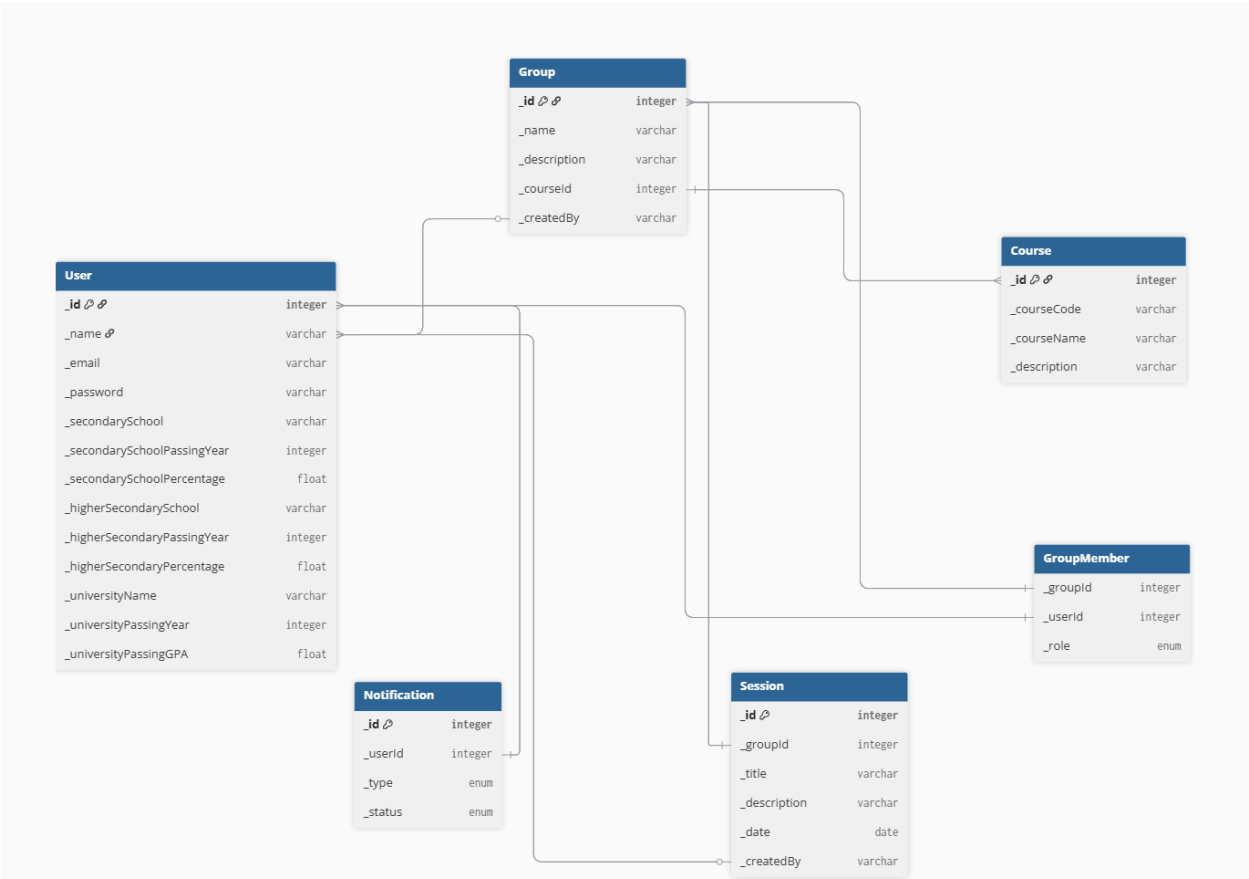
- Group calendar for scheduling study sessions.
- Session creation with title, description, date, and time.
- Email/push notifications for session reminders.

Outcomes:

- Calendar widget
- Event creation form
- Notification option

Database schema:

1. **User**
    a. **_id:** Unique identifier for the user
    b. **_name:** Name of the user
    c. **_email:** Email of the user
    d. **_password:** Password
    e. **_secondarySchool:** Name of secondary school
    f. **_secondarySchoolPassingYear:** Passing year from secondary school
    g. **_secondarySchoolPercentage:** Passing percentage of secondary school
    h. **_higherSecondarySchool:** Name of higher secondary school
    i. **_higherSecondaryPassingYear:** Passing year of higher secondary school
    j. **_higherSecondaryPercentage:** Passing percentage of higher secondary school
    k. **_universityName:** Name of the university
    l. **_universityPassingYear:** Passing year of university
    m. **_universityPassingGPA:** Passing GPA of the university
2. **Course**
    a. **_id:** Course id
    b. **_courseCode:** Unique identifier for the course
    c. **_courseName:** Name of the course
    d. **_description:** Course description
3. **Session**
    a. **_id:** Unique identifier of session
    b. **_groupId:** Id of the group to which the session belongs
    c. **_title:** Title of the session
    d. **_description:** Session description
    e. **_date:** Date of session.
    f. **_createdBy:** Name of user who created the session
4. **Notification**
    a. **_id:** Unique identifier of notification
    b. **_userId:** Id of the user
    c. **_type:** [Reminder, Invitation]
    d. **_status:** [Read, Unread]

5. **Group**
    a. **_id**: Unique identifier of the group
    b. **_name**: Group name
    c. **_description**: Group description
    d. **_courseId**: Course id of the group
    e. **_createdBy**: Name of the user creating the group
6. **GroupMember**
    a. **_groupId:** Id of the group
    b. **_userId:** Id of the user
    c. **_role:** [Admin, Member]

Database diagram: